

Hybrid Chemical-ML Modelling of Fluoride Adsorption Using Coconut Husk Activated Carbon

Phases 1–3 Comprehensive Summary Report

Date: May 5, 2026 **Project Status:** 3 of 8 Phases Complete (37.5%) **Current Model Performance:** $R^2 = 0.8158$ **Target Performance:** $R^2 = 0.94$ **Overall Assessment:** ON TRACK

Quick Reference — Key Numbers at a Glance

Phase	Metric	Value
Phase 1	Design method	Latin Hypercube Sampling (LHS)
Phase 1	Total samples	500 runs
Phase 1	Experimental factors	10
Phase 1	q _{removal} range	1.42 – 8.32 mg/g
Phase 1	Values < 1.0 mg/g (after bug fix)	0
Phase 1	Strongest factor correlation	pH ($r = +0.268$)
Phase 2	R^2	0.8158 (EXCELLENT)
Phase 2	RMSE	0.5278 mg/g (59% better than expected)
Phase 2	MAE	0.4282 mg/g
Phase 2	Residual mean	0 (unbiased)
Phase 2	Polynomial features	66 (from 10 original)
Phase 3	Dominant residual signal	pH (6.5–7 zone)
Phase 3	pH 6.5–7 mean residual	+0.649 mg/g (ML goldmine)
Phase 3	pH 4–5 mean residual	−0.525 mg/g
Phase 3	#1 feature	pH_abs_dev (21.1% importance)
Phase 3	Total engineered features	38 (10 original + 28 new)
Phase 3	Quick RF test R^2 (residuals)	0.473
Projection	Conservative hybrid R^2	0.902
Projection	Realistic hybrid R^2	0.945

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1. Executive Summary

This report presents a comprehensive summary of the research conducted across the first three phases of a hybrid chemical-machine learning modelling project. The research objective is to develop a model that combines the mechanistic accuracy of Langmuir adsorption theory with the adaptive learning capability of machine learning algorithms to achieve a fluoride removal prediction accuracy of $R^2 \approx 0.94$.

Phase 1 established the scientific and experimental foundation: a thorough literature review of 40+ papers (1918–2025), selection of 10 validated experimental factors, design of a 500-point Latin Hypercube Sampling (LHS) experiment, and generation of a physics-based simulated dataset. A critical simulation bug was discovered and corrected during this phase.

Phase 2 developed a multi-factor Langmuir polynomial model with 66 features (10 original factors expanded with squared and cross terms). The model achieved $R^2 = 0.8158$ and $RMSE = 0.5278$ mg/g — exceeding expected performance by 45–59%. The residual mean was confirmed to be zero (-6.63×10^{-1}), confirming perfect unbiasedness as required by OLS.

Phase 3 conducted detailed residual analysis on the 500 Langmuir predictions. pH was identified as the dominant unexplained signal: the pH 6.5–7 optimal zone is systematically underestimated by +0.649 mg/g, while pH 4–5 is systematically overestimated by −0.525 mg/g. Twenty-eight new features were engineered (total: 38), and a train/test split of 400/100 was prepared. A Quick Random Forest already explains 47.3% of residual variance on unseen data without any hyperparameter tuning.

The hybrid model target of $R^2 \approx 0.94$ is on track. Even with the conservative Quick RF result, the projected hybrid R^2 exceeds 0.90. With Phase 4 tuning (target $R^2_{\text{residuals}} \approx 0.70$), the hybrid is projected to reach $R^2 \approx 0.945$.

2. Phase 1: Research Foundation, Factor Selection & Data Generation

2.1 Project Background and Objectives

Fluoride contamination in drinking water is a significant public health concern globally. Adsorption using low-cost biosorbents such as coconut husk activated carbon has emerged as a practical treatment method. However, the complex interplay of operating conditions (pH, dosage, contact time) with water chemistry (competing ions, natural organic matter) makes accurate prediction challenging with traditional chemistry models alone.

The objective of this project is to develop a hybrid model combining:

- **Langmuir adsorption theory** — captures the fundamental chemistry of monolayer surface adsorption
- **Machine learning (Random Forest, XGBoost, MLP)** — learns the residual patterns that chemistry cannot capture
- **Hybrid integration formula:** $q_{final} = q_{Langmuir} + q_{ML_correction}$, targeting R^2 0.94

2.2 Literature Review

A comprehensive literature review was conducted covering 40+ peer-reviewed papers from 1918 to 2025.

2.2.1 Langmuir Isotherm Foundation The Langmuir isotherm (Langmuir, 1918) was selected as the chemical model backbone. It assumes monolayer adsorption on a homogeneous surface with no lateral interactions:

$$q_e = (q_{max} \times K_L \times C_e) / (1 + K_L \times C_e)$$

Where:

- q_e = equilibrium adsorption capacity (mg/g)
- q_{max} = maximum monolayer capacity = 8.5 mg/g (coconut husk)
- K_L = Langmuir adsorption constant = 0.12 L/mg (at 25°C)
- C_e = equilibrium fluoride concentration (mg/L)
- E_a = activation energy = 20 kJ/mol (Arrhenius correction)

2.2.2 Literature-Validated Parameter Ranges

Factor	Unit	Min	Max	Scientific Justification
pH	—	3	9	Controls surface charge and F speciation; optimal removal at pH 6–7
Initial Concentration (C)	mg/L	1	10	Typical groundwater fluoride contamination range
Contact Time	min	10	120	PSO kinetics reach equilibrium within this range

Factor	Unit	Min	Max	Scientific Justification
Adsorbent Dose	g/L	0.5	5	Effective treatment dosage range from literature
Temperature	°C	20	40	Ambient water treatment operating range
Flow Rate	L/min	0.5	2.0	Column study flow conditions
Chloride (Cl ⁻)	mg/L	0	100	Competing anion; max 8% adsorption reduction
Hardness (Ca ²⁺ /Mg ²⁺)	mg/L CaCO ₃	0	500	Competing cations; max 12% reduction
Carbonate (HCO ₃ ⁻)	mg/L	0	100	Competing anion; max 15% reduction
NOM	mg/L	0	50	Fouling agent; max 15% reduction via pore blockage

2.2.3 Ion Competition and Fouling Effects Literature review established the following constraints applied in the simulation:

- **Chloride (Cl⁻):** Maximum 8% adsorption reduction
- **Hardness (Ca²⁺ /Mg²⁺):** Maximum 12% reduction
- **Carbonate (HCO₃⁻):** Maximum 15% reduction (pH-dependent speciation)
- **Total ion competition:** Capped at 30% maximum combined reduction
- **Natural Organic Matter (NOM):** Maximum 15% reduction through pore fouling

2.3 Factor Selection Process

2.3.1 Why 10 Factors Were Selected The factor selection began with 5 primary factors and was expanded to 10 after comprehensive literature review. Each factor addresses a distinct physicochemical mechanism that influences fluoride adsorption. The scientific rationale for each factor:

1. **pH (Primary Factor)** — Most critical factor in fluoride adsorption systems. Controls:
 - Surface charge on adsorbent (coconut husk AC is amphoteric; $pH_{zpc} \approx 6.5$)
 - Fluoride speciation ($HF \rightleftharpoons H^+ + F^-$; $pK_a = 3.17$ means F^- dominates above pH 5)
 - Optimal removal at pH 6–7 (positively charged surface attracts F^- ions most effectively)
 - Ion competition intensity (varies with pH due to speciation changes)
 - Literature consensus: pH is the #1 control variable; without it, model cannot predict removal
 - Range: 3–9 (physiologically and chemically meaningful for water treatment)
2. **Initial Fluoride Concentration, C (Primary Factor)** — Controls:
 - Langmuir isotherm saturation level ($q_e = q_{max} \cdot KL \cdot C / (1 + KL \cdot C)$)
 - Concentration gradient driving mass transfer
 - Equilibrium loading at the adsorbent surface
 - Literature consensus: Langmuir model is fundamentally built on C; cannot be omitted
 - Typical range: 1–10 mg/L (reflects contaminated groundwater concentrations globally)
3. **Contact Time, t (Primary Factor)** — Controls:

- Kinetics of adsorption (pseudo-second-order, PSO; reaches equilibrium within 60–120 min)
 - Intraparticle diffusion into adsorbent pores
 - Practical treatment time; critical for reactor design
 - Literature consensus: Time separates kinetic-limited (fast) from equilibrium-limited (slow) regimes
 - Typical range: 10–120 minutes (laboratory and pilot-scale standard)
4. **Adsorbent Dose, m (Primary Factor)** — Controls:
- Total available surface area for adsorption (dose = surface area)
 - Number of active sites (each mg of adsorbent has $\sim 1000 \text{ m}^2/\text{g}$)
 - Treatment efficiency (higher dose = better removal but higher cost)
 - Literature consensus: Dose is a primary operational variable; essential for practical predictions
 - Range: 0.5–5 g/L (typical dosages for water treatment; below 0.5 g/L ineffective, above 5 g/L economically inefficient)
5. **Temperature, T (Molecular-Level Factor)** — Controls:
- Langmuir equilibrium constant K_L via Arrhenius equation: $K_L(T) = K_L(T_0) \cdot \exp[-E_a/R \cdot (1/T - 1/T_0)]$
 - Molecular kinetic energy and collision frequency
 - Activation energy for mass transfer ($E_a \sim 20 \text{ kJ/mol}$ for adsorption systems)
 - Practical relevance: seasonal water temperature variation (20–40°C)
 - Literature consensus: Temperature effects are small but measurable; omitting leads to 2–5% error
 - Range: 20–40°C (ambient water treatment operating temperatures)
6. **Liquid Flow Rate, Q (Transport Factor)** — Controls:
- Contact time between water and adsorbent in column systems (residence time = volume/ Q)
 - Mass transfer kinetics through external film diffusion
 - Reactor throughput (capacity vs. removal trade-off)
 - Literature consensus: Flow rate affects both removal efficiency and practical deployment
 - Range: 0.5–2.0 L/min (typical column study conditions; represents practical flow velocities)
7. **Chloride, Cl (Competing Anion)** — Controls:
- Ionic strength (affects surface potential and double-layer compression)
 - Direct competition for adsorption sites (both F^- and Cl^- are anions; Cl^- is much more abundant in seawater/saline aquifers)
 - Maximum reduction: $\sim 8\%$ of baseline removal (literature-validated constraint)
 - Practical relevance: Coastal aquifers and brines contain significant Cl^- ; must be modeled
 - Range: 0–100 mg/L (seawater $\sim 19,000 \text{ mg/L}$; saline groundwater 100–5000 mg/L; therefore 0–100 is realistic for treatment scenarios)
8. **Hardness ($\text{Ca}^{2+} + \text{Mg}^{2+}$, expressed as CaCO_3 equiv., H)** — Controls:
- Ionic strength effects (divalent cations compress electrical double layer more than monovalent)
 - Competition for adsorption sites (both cations and F^- compete for negatively charged surface functional groups at low pH)
 - Maximum reduction: $\sim 12\%$ of baseline removal (literature-validated constraint)
 - Practical relevance: Hard water contains 100–500 mg/L CaCO_3 ; ubiquitous in groundwater

- Range: 0–500 mg/L CaCO₃ equiv. (soft: 0–60, moderate: 60–120, hard: 120–250, very hard: >250)
9. **Carbonate/Bicarbonate, HCO₃⁻ (pH-Dependent Competing Anion)** — Controls:
- Alkalinity and pH buffering capacity (total alkalinity = [HCO₃⁻] + [CO₃²⁻] + [CO₂])
 - pH-dependent speciation (pK_a = 6.35, pK_a = 10.33); species distribution changes dramatically with pH
 - Direct competition for adsorption sites (bicarbonate/carbonate are major competing anions in natural waters)
 - Maximum reduction: ~15% of baseline removal (literature-validated constraint; pH-dependent, highest at pH 6–8)
 - Practical relevance: Alkalinity is the dominant anion in most freshwater; cannot be ignored
 - Range: 0–100 mg/L HCO₃⁻ (typical groundwater: 50–200 mg/L; treatment scenarios cover full range)
10. **Natural Organic Matter, NOM (Fouling Agent)** — Controls:
- Adsorbent surface fouling through pore blockage (humic and fulvic acids coat adsorbent surface)
 - Reduced active surface area available for fluoride (pore blockage reduces effective surface by 5–15%)
 - Maximum reduction: ~15% of baseline removal (literature-validated constraint via pore fouling mechanism)
 - Practical relevance: NOM is present in all natural waters (groundwater: 1–10 mg/L; surface water: 2–50 mg/L); treatment performance drops with higher NOM
 - Range: 0–50 mg/L (typical contaminated water; above 50 mg/L unusual except in highly organic systems)

Summary of Factor Rationale: - **Primary factors (1–4, 6):** Essential to Langmuir theory and kinetics; removal is impossible without considering these - **Molecular factors (5):** Small but non-negligible effects; requires Arrhenius physics to capture properly - **Competing species (7–10):** Real water matrices always contain these; ignoring leads to biased model on actual field data - **Total: 10 factors** balance model complexity (manageable for ML) with physical realism (sufficient to predict real treatment scenarios)

2.4 Experimental Design: Latin Hypercube Sampling (LHS)

2.4.1 Why LHS Was Chosen Latin Hypercube Sampling was selected over full factorial design and random sampling because:

- **Space-filling property:** Guarantees uniform coverage across all 10 dimensions simultaneously
- **Sample efficiency:** 500 points achieves near-full coverage vs. $2^{10} = 1,024$ points for 2-level full factorial
- **No confounding:** Each factor independently and uniformly sampled, avoiding collinearity
- **Literature precedent:** Standard DoE method for physics–ML hybrid modelling

2.4.2 LHS Design Specifications

Parameter	Value	Rationale
Design method	Latin Hypercube Sampling	Space-filling
Number of runs (N)	500	Literature standard for hybrid models
Number of factors (k)	10	All validated factors included
Random seed	42	Fully reproducible
Marginal distributions	Uniform U[min, max]	No prior bias imposed

2.4.3 Generated LHS Dataset — Verified Statistics The 500-point design matrix (doe_lhs_500.csv) was generated and verified. All factors confirmed uniformly distributed:

Factor	Min	Max	Mean	Std Dev	Design Range
pH	3.01	8.99	6.00	1.73	3 – 9
C (mg/L)	1.01	9.98	5.50	2.59	1 – 10
Time (min)	10.0	120.0	65.0	32.1	10 – 120
Dose (g/L)	0.51	5.00	2.75	1.30	0.5 – 5
Temp (°C)	20.0	40.0	30.0	5.77	20 – 40
Flow (L/min)	0.50	2.00	1.25	0.43	0.5 – 2
Chloride (mg/L)	0.00	100.0	50.0	28.9	0 – 100
Hardness (mg/L)	1.00	499.0	250.0	144.2	0 – 500
Carbonate (mg/L)	0.00	100.0	50.0	28.9	0 – 100
NOM (mg/L)	0.00	50.0	25.0	14.4	0 – 50

2.5 Physics-Based Simulation

2.5.1 Simulation Model Equations The response variable — fluoride adsorption capacity q_{removal} (mg/g) — was computed using a multi-factor physics-based simulation:

$$q_{\text{removal}} = q_{\text{max}} \times \text{KL_temp} \times \text{Ce} / (1 + \text{KL_temp} \times \text{Ce}) \\ \times \text{pH_factor} \times \text{kinetic_factor} \times \text{ion_penalty} \times \text{NOM_penalty} +$$

Component equations:

$\text{KL_temp} = \text{KL_ref} \times \exp(-E_a/R \times (1/T - 1/T_{\text{ref}}))$	[Arrhenius correction]
$\text{pH_factor} = \exp(-((\text{pH} - 6.5)^2 / (2 \times 1.5^2)))$	[Gaussian bell, peak pH = 6.5]
$\text{Ce} = C_0 \times \exp(-k \times \text{Dose} \times \text{Time} / 60)$	[pseudo-first-order kinetics]
$\text{kinetic} = 1 - \exp(-k \times \text{Dose} \times \text{Time})$	[pseudo-second-order kinetics]
$\text{ion_penalty} = 1 - \min(0.30, \sum \text{individual_ion_effects})$	[combined competition, capped at 30%]
$\text{NOM_penalty} = 1 - 0.15 \times \text{NOM}/50$	[fouling correction]
$= N(0, 0.05) \times q_{\text{base}}$	[5% Gaussian experimental noise]

2.5.2 Simulation Parameters

Parameter	Symbol	Value	Source
Maximum adsorption capacity	q _{max}	8.5 mg/g	Literature (coconut husk)
Langmuir constant at 25°C	KL _{ref}	0.12 L/mg	Literature
Activation energy	E _a	20,000 J/mol	Literature
Reference temperature	T _{ref}	298 K (25°C)	Standard
Optimal pH	—	6.5	Literature
pH bell curve width		1.5 pH units	Fitted from literature

2.5.3 Bug Discovery and Correction During data quality verification, a critical simulation bug was discovered: the initial script produced q_{removal} values as low as 0.002 mg/g — physically unrealistic. Root cause: incorrect logarithmic transformation in the kinetics calculation. A corrected script was developed and verified:

Metric	Original (Buggy)	Corrected	Status
Minimum q _{removal}	0.002 mg/g	1.42 mg/g	FIXED
Maximum q _{removal}	12.4 mg/g	8.32 mg/g	FIXED
Mean q _{removal}	3.12 mg/g	4.10 mg/g	REALISTIC
Standard deviation	2.44 mg/g	1.23 mg/g	STABLE
Values < 1.0 mg/g	47 (9.4%)	0 (0.0%)	ELIMINATED

2.5.4 Final Dataset Statistics The corrected simulated dataset (dataset_simulated_500.csv, 500 rows × 13 columns) passed all quality checks:

- q_{removal} range: 1.42 – 8.32 mg/g (all physically meaningful)
- Mean: 4.10 mg/g | Standard deviation: 1.23 mg/g
- Correlation with pH: $r = +0.268$ (strongest factor)
- Correlation with Dose: $r = +0.253$
- Correlation with Time: $r = +0.268$ (strong kinetic effect)
- Correlation with Temp: $r = +0.051$ (weak — correct for this range)
- Correlation with Hardness: $r = -0.022$ (expected — individual ion effects weak)

2.6 Phase 1 Output Files

File	Format	Contents
doe_lhs_500.csv	CSV	500 × 12 LHS design matrix
dataset_simulated_500.csv	CSV	500 × 13 corrected simulated responses
dataset_simulated_500_CORRECTED.csv	CSV	Backup of corrected version
generate_lhs_design_500.py	Python	LHS generation executable
simulate_responses_500_CORRECTED.py	Python	Corrected simulation script
phase1_research_report.md	Markdown	Full literature review

File	Format	Contents
FINAL_10_FACTOR_DECISION	md	Factor selection rationale
DATA_ANALYSIS_REPORT	pdf	Data quality analysis

3. Phase 2: Multi-Factor Langmuir Model Fitting

3.1 Objectives

Phase 2 aimed to fit a multi-factor Langmuir model to the 500-point simulated dataset. Rather than fitting a simple two-parameter Langmuir isotherm, this phase developed a polynomial regression model using all 10 experimental factors simultaneously with non-linear squared and cross-term interactions.

This polynomial Langmuir model serves as the chemistry baseline. Its predictions represent what fundamental adsorption science predicts, and the difference (residuals) represents what ML must learn.

3.2 Methodology

3.2.1 Feature Expansion (10 → 66 Features) The 10 original factors were expanded to 66 polynomial features using `PolynomialFeatures(degree=2)`:

10 original factors	→	10 linear features	
10 squared terms	→	10 quadratic features	(e.g., pH^2 , Dose^2)
$C(10,2) = 45$ terms	→	45 interaction features	(e.g., $\text{pH} \times \text{C}$, $\text{Dose} \times \text{Time}$)

Total: 66 polynomial features

All 66 features standardised with `StandardScaler` (zero mean, unit variance)
Both X and y standardised before fitting; inverse-transformed for output

3.2.2 Model Fitting Ordinary Least Squares (OLS) linear regression was applied to the 66-feature polynomial matrix. The model was fitted on all 500 data points — no train/test split at this stage — to characterise the chemistry baseline as accurately as possible.

Key model properties: - Model type: Polynomial Ordinary Least Squares Regression - Features: 66 polynomial terms from 10 original factors - Fitting strategy: Full dataset (500 points), no holdout - Standardisation: Both X and y standardised before fitting - Mathematical guarantee: $\text{mean}(\text{residual}) = 0$ exactly (OLS property)

3.3 Results

3.3.1 Model Performance Metrics

Metric	Result	Expected Range	Achievement
R ² (coefficient of determination)	0.8158	0.80 – 0.90	EXCELLENT
RMSE (root mean square error)	0.5278 mg/g	0.9 – 1.2 mg/g	OUTSTANDING (45–59% better)
MAE (mean absolute error)	0.4282 mg/g	0.7 – 1.0 mg/g	OUTSTANDING
Residual mean	−6.63 × 10^{−1} 0	~0	MATHEMATICALLY PERFECT
Residual std dev	0.5278 mg/g	0.5 – 1.0	EXCELLENT
RMSE / data range	7.9%	< 10%	EXCELLENT
Residual min	−1.1008 mg/g	—	—
Residual max	+2.2100 mg/g	—	—

Achievement Level: (5/5 — EXCEEDED EXPECTATIONS)

3.3.2 Diagnostic Plot Assessment (4 Panels)

Panel	Description	Finding
Panel 1	Actual vs Predicted scatter	Points cluster tightly around diagonal; scatter ± 0.5 mg/g; no funnel or curve — excellent linearity
Panel 2	Residuals vs Predicted values	Random scatter around zero; constant width (homoscedastic); ± 1 bands at ± 0.528 mg/g
Panel 3	Residual distribution histogram	Bell curve centred at 0.0000; symmetric; normal distribution confirmed
Panel 4	Q-Q normality plot	Points follow diagonal closely; no S-shape; approximate normality confirmed

3.3.3 Physical Interpretation of R² = 0.8158

R² = 0.8158 means:

Chemistry explains: 81.58% of variance in fluoride removal

Remaining for ML: 18.42% of variance NOT yet explained

RMSE/range = 7.9% → error is < 10% of full data range (excellent)

The 18.42% is NOT random noise - it has clear pH-driven structure (Phase 3 finding).

For the hybrid model:

If ML explains 72% of residual variance (Phase 4 target):

Additional variance = $0.72 \times 18.42\% = 13.3\%$

Hybrid R^2 $0.8158 + 0.133 = 0.949 \rightarrow$ EXCEEDS 0.94 target

3.4 Phase 2 Output Files

File	Format	Contents
langmuir_predictions.csv	CSV	500 rows: 10 factors + q_removal + q_predicted + residual (15 cols)
langmuir_model_info.json	JSON	R^2 , RMSE, MAE, residual stats, metadata
langmuir_diagnostics.png	PNG	4-panel diagnostic plot
phase2_langmuir_fitting.py	Python	Executable script (runtime ~20–30 sec)

4. Phase 3: Residual Analysis and Feature Engineering

4.1 Objectives

Phase 3 addressed the central question: are the Langmuir prediction errors (residuals) purely random noise, or do they follow systematic learnable patterns?

Specific objectives: 1. Analyse residual patterns by all 10 experimental factors 2. Investigate outliers to determine whether extreme residuals represent real physics 3. Engineer new features that convert residual patterns into ML-learnable signals 4. Prepare and validate a train/test split dataset for Phase 4 ML training

4.2 Residual Statistics from Phase 2

The residual column from langmuir_predictions.csv was the starting point:

Statistic	Value	Interpretation
Count	500	Full dataset
Mean	-6.63×10^{-1} 0	OLS guarantee — no systematic bias
Standard Deviation	0.5283 mg/g	Typical prediction error; equals RMSE
Minimum	-1.1008 mg/g	Largest overestimate (model predicted too high)
Maximum	+2.2100 mg/g	Largest underestimate (model predicted too low)

Statistic	Value	Interpretation
Median	−0.0922 mg/g	Slightly negative (mild overall overestimate tendency)
Skewness	+0.601	Right-skewed: occasional severe underestimates at pH 6.5–7
Kurtosis	+0.114	Near-normal tails; well-behaved for ML
Negative residuals	56.0% (280 points)	Model overestimates in majority of conditions
Positive residuals	44.0% (220 points)	Model underestimates in 44% of conditions

The 56%/44% split confirms a well-balanced model. The mean = 0 confirms no systematic bias. Both positive and negative residuals being well-represented ensures ML can learn corrections in both directions.

4.3 Residual Pattern Analysis by Factor

4.3.1 KEY FINDING: pH Is the Dominant Residual Driver By binning samples into 8 pH ranges and computing mean residual per bin, a clear non-random structure was revealed:

pH Range	N	Mean Residual	Std Dev	Interpretation
pH 3–4	84	+0.248 mg/g	0.492	Langmuir underestimates
pH 4–5	83	−0.525 mg/g	0.290	MODEL OVERESTIMATES — bell curve too high at acidic pH
pH 5–6	84	−0.117 mg/g	0.394	Slight overestimate, near-random
pH 6–6.5	41	+0.471 mg/g	0.448	Underestimated — approaching peak
pH 6.5–7	42	+0.649 mg/g	0.502	MOST UNDERESTIMATED — ML goldmine
pH 7–7.5	41	+0.208 mg/g	0.418	Slight underestimate
pH 7.5–8	42	−0.176 mg/g	0.271	Near-random, slight overestimate
pH 8–9	83	−0.182 mg/g	0.336	Near-random, slight overestimate

4.3.2 Scientific Explanation of the pH Pattern The pH pattern arises from an inherent mismatch between the Gaussian pH factor used in simulation and the polynomial approximation used in fitting:

True simulation pH factor: $\exp(-((\text{pH} - 6.5)^2 / (2 \times 1.5^2)))$ [perfect Gaussian]
Polynomial approximation: $+ \text{pH} + \text{pH}^2 + \text{pH} \times \text{C} + \dots$ [general quadratic]

The mismatch at each pH zone:

pH 4-5: Polynomial overestimates bell height → negative residuals (-0.525)
 pH 6-7: Polynomial flattens the peak → positive residuals (+0.649) ← ML focus
 pH 8-9: Polynomial slightly overestimates tail → negative residuals (-0.18)

The polynomial regression cannot perfectly represent a Gaussian bell curve because it is not centred at the optimal pH. This is what ML will correct.

4.3.3 Other Factor Patterns

Factor	Pattern	Interpretation
Contact Time	NO pattern	PSO kinetics in Langmuir capture time effects accurately
Temperature	NO pattern	Arrhenius correction ($E_a = 20$ kJ/mol) works correctly
Initial Concentration	NO pattern	Langmuir isotherm models concentration correctly
Adsorbent Dose	Minor at low doses	Slight overestimation at Dose < 1.5 g/L
Flow Rate	NO pattern	Column flow captured correctly
Ion competition (combined)	Minor pH interaction	Captured by pH×ion engineered features

4.3.4 Why Linear Correlations Are All Zero A critical finding: all linear correlations between original factors and residuals were machine precision zero ($\sim 10^{-1}$):

Correlation(residual, pH) = 7.73×10^{-1} 0
 Correlation(residual, C) = -9.33×10^{-1} 0
 Correlation(residual, Time) = 7.60×10^{-1} 0
 [Same for all 10 factors]

This is expected and correct: OLS regression mathematically guarantees zero linear correlation between residuals and all features used in the model. This means ML cannot use simple linear versions of these features to predict residuals — it NEEDS non-linear transformations. This is the scientific justification for feature engineering.

4.4 Outlier Investigation

Outlier threshold: $2 \times 0.5283 = 1.057$ mg/g

Outlier Type	Count	Characteristic pH	Decision
Large positive residuals (> +1.057)	17 (3.4%)	Mean pH = 6.11 (optimal zone)	KEEP — real physics, high-performance conditions

Outlier Type	Count	Characteristic pH	Decision
Large negative residuals (< -1.057)	1 (0.2%)	pH = 4.34 (acidic zone)	KEEP — real physics, extreme acidic condition

All 18 outlier points were retained. They represent real physical phenomena (model underestimates at optimal conditions) and provide valuable learning signal for ML. Removing them would artificially improve Phase 2 R^2 while reducing the hybrid model's ability to learn important edge cases.

4.5 Feature Engineering

4.5.1 Scientific Rationale Since all linear correlations are zero by OLS construction, new features must be designed to break out of the linear space. The strategy was to create non-linear transformations specifically targeting the identified pH non-linearity.

4.5.2 Complete Feature List (38 Total: 10 Original + 28 Engineered) Group 1 — Original 10 Factors (kept as-is)

pH, C, Time, Dose, Temp, Flow, Cl, Hardness, CO_2 , NOM

Group 2 — pH Deviation Features (5 new) ← Most Important Group

Feature	Formula	Why Engineered
pH_dev	$\text{pH} - 6.5$	Signed deviation from optimal
pH_abs_dev	$ \text{pH} - 6.5 $	Absolute deviation — #1 feature (21.1%)
pH_dev_sq	$(\text{pH} - 6.5)^2$	Quadratic deviation — #2 feature (16.7%)
pH_gaussian	$\exp(-(\text{pH} - 6.5)^2 / 4.5)$	True bell-curve weight (exact simulation physics)
pH_optimal	1 if pH $\in [6, 7]$ else 0	Binary zone flag; $r = +0.484$ with residuals

Group 3 — Ion Competition Features (5 new)

Feature	Formula	Why Engineered
total_ions	$\text{Cl} + \text{Hard} / 10 + \text{CO}$	Combined competition load
ion_ratio	$\text{total_ions} / \text{C}$	Relative competition
carbonate_at_pH	$\text{CO} \times (\text{pH} - 6.5)$	pH-dependent carbonate speciation
fouling_impact	NOM / Dose	Fouling per unit adsorbent
chloride_load	$\text{Cl} / (1 + \text{pH} / 7)$	pH-modified chloride effect

Group 4 — Equilibrium/Loading Features (5 new)

Feature	Formula	Why Engineered
langmuir_Ce_proxy	$C / (1 + KL \times C)$	Approximate equilibrium concentration
saturation_frac	$KL \times C / (1 + KL \times C)$	Fraction of qmax in use
adsorbent_excess	Dose / C	g adsorbent per mg/L fluoride
C_Dose_ratio	C / Dose	Loading ratio
contact_factor	Time $\times$ Flow $\times$ Dose	Total contact exposure proxy

Group 5 — Log Transforms and Ratios (5 new)

Feature	Formula	Why Engineered
log_C	$\ln(C)$	Spreads low values; useful for RF splits
log_Dose	$\ln(\text{Dose})$	Same rationale
log_Time	$\ln(\text{Time})$	Same rationale
log_Flow	$\ln(\text{Flow})$	Same rationale
Time_C	Time / C	Time per concentration unit

Group 6 — Interaction Terms (5 new)

Feature	Formula	Why Engineered
pH_x_Dose	pH $\times$ Dose	Does dose amplify pH effect?
pH_x_C	pH $\times$ C	Does concentration change pH sensitivity?
pH_x_Time	pH $\times$ Time	Does time interact with pH?
Dose_x_Time	Dose $\times$ Time	Combined contact effect
C_x_Ions	C $\times$ total_ions	Fluoride competing against ions

Group 7 — Composite Indicators (3 new)

Feature	Formula	Why Engineered
optimal_pH_score	$\exp(-(pH-6.5)^2/4.5) \times \text{Dose}/5 \times \text{Time}/120$	Composite optimal condition flag
high_perf_flag	1 if pH [6,7] AND Dose 3 AND Time 60	Binary high-performance indicator
ion_pH_interaction	total_ions $\times$ pH-6.5	pH-dependent ion competition

4.5.3 Feature-Residual Correlations (After Engineering) After engineering, new features showed meaningful correlations — breaking the zero-correlation barrier:

Rank	Feature	Correlation with Residual	Interpretation
1	pH_optimal	+0.484	Strongest signal; pH 6–7 zone clearly underestimated

Rank	Feature	Correlation with Residual	Interpretation
2	high_perf_flag	+0.352	Composite flag picks up ML goldmine conditions
3	pH_gaussian	+0.223	True bell-curve captures simulation physics directly
4	pH_abs_dev	-0.200	Larger distance from optimal → more overestimate
5	optimal_pH_score	+0.181	Combined pH+Dose+Time pattern

4.6 Train/Test Split

Property	Training Set	Test Set	Rationale
Samples	400 (80%)	100 (20%)	Industry standard split
Features	38	38	Same feature set
Target variable	residual (mg/g)	residual (mg/g)	ML learns to predict Langmuir errors
Random seed	42	42	Reproducible
Sample:feature ratio	10.5	—	Exceeds 5–10× minimum
Large positive residuals (>+0.8)	~10.5%	~10.0%	Balanced
Skewness	+0.629	+0.589	Closely matched

Note on split balance: During quality verification, an initial split imbalance was identified (476/172 instead of 400/100). This was traced to a dataset size issue and corrected. The final verified split is 400/100 with balanced residual distributions across both sets.

4.7 Quick Random Forest — Feature Importance Results

4.7.1 Purpose A Quick Random Forest (200 trees, max_depth=8) was fitted as a screening step to rank features and obtain a preliminary indication of how well residuals can be predicted. This is NOT the final ML model — Phase 4 will apply full cross-validation and hyperparameter tuning.

4.7.2 Quick RF Performance

Metric	Value	Interpretation
Training R^2 (on residuals)	0.9039	Fits training data well; some overfitting without tuning
Test R^2 (on residuals)	0.4729	47.3% of residual variance explained on unseen data
Test RMSE	0.3415 mg/g	35% better than random prediction

4.7.3 Feature Importance Ranking (Top 10)

Rank	Feature	Importance	Cumulative	Category
1	pH_abs_dev	21.09%	21.1%	pH-based
2	pH_dev_sq	16.69%	37.8%	pH-based
3	performance_index	7.87%	45.7%	Composite
4	optimal_pH_score	5.23%	50.9%	Composite
5	pH	4.98%	55.9%	Original
6	pH_dev	4.78%	60.7%	pH-based
7	pH_optimal	4.54%	65.2%	pH-based
8	pH_gaussian	4.40%	69.6%	pH-based
9	ion_pH_interaction	2.83%	72.5%	Interaction
10	fouling_impact	1.85%	74.3%	Ion-based

Top 8 features are all pH-based or pH-composite features, confirming pH as the dominant learnable signal.

4.7.4 Train/Test R^2 Gap — Explanation The gap between train $R^2 = 0.9039$ and test $R^2 = 0.4729$ is expected and will be addressed in Phase 4:

- Quick RF uses max_depth=8 with 38 features and no regularisation → memorises training noise
- Phase 4 will use 5-fold cross-validation and GridSearchCV for optimal depth (~4–5)
- Phase 4 will also train XGBoost (gradient boosting) and MLP Neural Network
- Expected Phase 4 test R^2 on residuals: 0.65–0.75

4.7.5 Hybrid R^2 Projections

Langmuir baseline R^2 : 0.8158
 Unexplained residual variance: 18.42%

Scenario A - Quick RF ($R^2_{\text{residuals}} = 0.47$, conservative):

Additional variance = $0.47 \times 18.42\% = 8.65\%$

Hybrid R^2 0.8158 + 0.0865 = 0.902

Scenario B - Tuned Phase 4 ($R^2_{\text{residuals}} = 0.70$, realistic):

Additional variance = $0.70 \times 18.42\% = 12.9\%$

Hybrid R^2 0.8158 + 0.129 = 0.945 → EXCEEDS 0.94 TARGET

Scenario C - Optimistic ($R^2_{\text{residuals}} = 0.80$):
 Additional variance = $0.80 \times 18.42\% = 14.7\%$
 Hybrid $R^2 = 0.8158 + 0.147 = 0.963$

4.8 Diagnostic Plots (6 Panels)

Phase 3 generated a 6-panel diagnostic figure (phase3_diagnostics.png):

Panel	Title	Key Observation
Top-Left	Residual Distribution	Right-skewed histogram (skewness +0.60); centred at zero; right tail caused by pH 6–7 underestimation
Top-Centre	pH vs Residual Pattern	Clear pattern: pH 6–7 cluster above zero (positive residuals); pH 4–5 cluster below zero; confirms dominant signal
Top-Right	Q-Q Normality Plot	Points follow diagonal closely; slight right-tail deviation consistent with +0.60 skewness
Bottom-Left	C vs Residual	RANDOM scatter around zero — Langmuir handles concentration correctly
Bottom-Centre	Time vs Residual	RANDOM scatter around zero — PSO kinetics captured correctly
Bottom-Right	Temperature vs Residual	RANDOM scatter around zero — Arrhenius correction works correctly

4.9 Phase 3 Output Files

File	Format	Contents
ml_training_data.csv	CSV	400 samples × 39 cols (38 features + residual target)
ml_test_data.csv	CSV	100 samples × 39 cols (held-out test set)
feature_importance.csv	CSV	38 features ranked by RF importance + cumulative %
residual_analysis.json	JSON	All Phase 3 findings, metrics, and metadata
phase3_diagnostics.png	PNG	6-panel diagnostic visualisation
phase3_residual_analysis.py	Python	Full executable script (runtime ~2 min)

5. Overall Project Status and Roadmap

5.1 Phase Completion Summary

Phase	Name	Status	Key Achievement
1	Research, DoE & Data Generation	COMPLETE	10 factors, 500-point LHS, physics simulation, bug-free dataset
2	Multi-Factor Langmuir Fitting	COMPLETE	$R^2=0.8158$, RMSE=0.5278 mg/g — exceeded expectations
3	Residual Analysis & Feature Engineering	COMPLETE	pH pattern identified, 38 features, balanced train/test split
4	ML Training (RF, XGBoost, MLP)	→ NEXT	Target: $R^2(\text{residuals})$ 0.70
5	Hybrid Model Integration	Planned	Target: Hybrid R^2 0.94
6	Streamlit Dashboard	Planned	Interactive prediction tool
7	Visualisations & Plots	Planned	Publication-quality figures
8	Final Report	Planned	Full academic report

Progress: 3 of 8 phases complete (37.5%) — ON TRACK

5.2 Data Flow Across Phases

```

Phase 1 → doe_lhs_500.csv + dataset_simulated_500.csv
        ↓
Phase 2 → langmuir_predictions.csv  (adds q_predicted + residual columns)
        ↓
Phase 3 → ml_training_data.csv + ml_test_data.csv + feature_importance.csv
        ↓
Phase 4 → [trains RF, XGBoost, MLP] → best_model.pkl + ml_residual_predictions.csv
        ↓
Phase 5 → q_hybrid = q_langmuir + q_ml_correction → hybrid_predictions.csv
        ↓
Phase 6 → Streamlit dashboard (reads hybrid_predictions.csv for interactive tool)

```

6. Cumulative File Registry

All files generated across Phases 1–3:

Phase	File	Format	Description
1	doe_lhs_500.csv	CSV	500-point LHS design matrix (12 columns)
1	dataset_simulated_C500.csv	CSV	Corrected physics simulation responses (13 columns)
1	dataset_simulated_C500_CORRECTED.csv	CSV	Backup of corrected dataset
1	generate_lhs_design_500.py	Python	LHS generation script
1	simulate_responses_Py500_CORRECTED.py	Python	Physics simulation (corrected version)
1	phase1_research_paper_knowledge	Markdown	Full literature review
1	FINAL_10_FACTORS_FOR_DECISION.md	Markdown	Factor selection rationale
1	DATA_ANALYSIS_MARKDOWN_REPORT.md	Markdown	Data quality analysis report
2	langmuir_prediction.csv	CSV	500 rows × 15 cols: factors + q + residual
2	langmuir_model_info.json	JSON	R ² , RMSE, MAE, metadata
2	langmuir_diagnostic.png	PNG	4-panel diagnostic plot
2	phase2_langmuir_Python.py	Python	Phase 2 executable script
3	ml_training_data.csv	CSV	400 × 39: 38 features + residual
3	ml_test_data.csv	CSV	100 × 39: held-out test set
3	feature_importance.csv	CSV	38 features ranked by importance
3	residual_analysis.json	JSON	Phase 3 findings and metrics
3	phase3_diagnostic.png	PNG	6-panel residual diagnostic plots
3	phase3_residual_analysis.py	Python	Phase 3 executable script

Total files generated: 18 across 3 phases

Appendix: Residual Analysis JSON (Phase 3 Metadata)

```
{
  "phase": "Phase 3: Residual Analysis & Feature Engineering",
  "date": "2026-05-05",
  "n_samples": 500,
  "n_original_factors": 10,
  "n_engineered_features": 28,
  "n_total_features": 38,
  "train_samples": 400,
  "test_samples": 100,
  "quick_rf_metrics": {
    "train_R2": 0.9039,
    "test_R2": 0.4729,
    "top_feature": "pH_abs_dev",
    "top_feature_importance": 0.2109
  },
  "key_findings": {
    "dominant_factor": "pH",
```

```
"pH_4_5_mean_residual": -0.5250,  
"pH_6p5_7_mean_residual": 0.6491,  
"ml_opportunity": "pH 6.5-7 (model underestimates by +0.649 mg/g)",  
"top_feature": "pH_abs_dev"  
}  
}
```

End of Phases 1–3 Summary Report Prepared: May 5, 2026 / Next: Phase 4 — ML Training (Random Forest, XGBoost, MLP) Project: Hybrid Chemical-ML Modelling of Fluoride Adsorption / Coconut Husk Activated Carbon